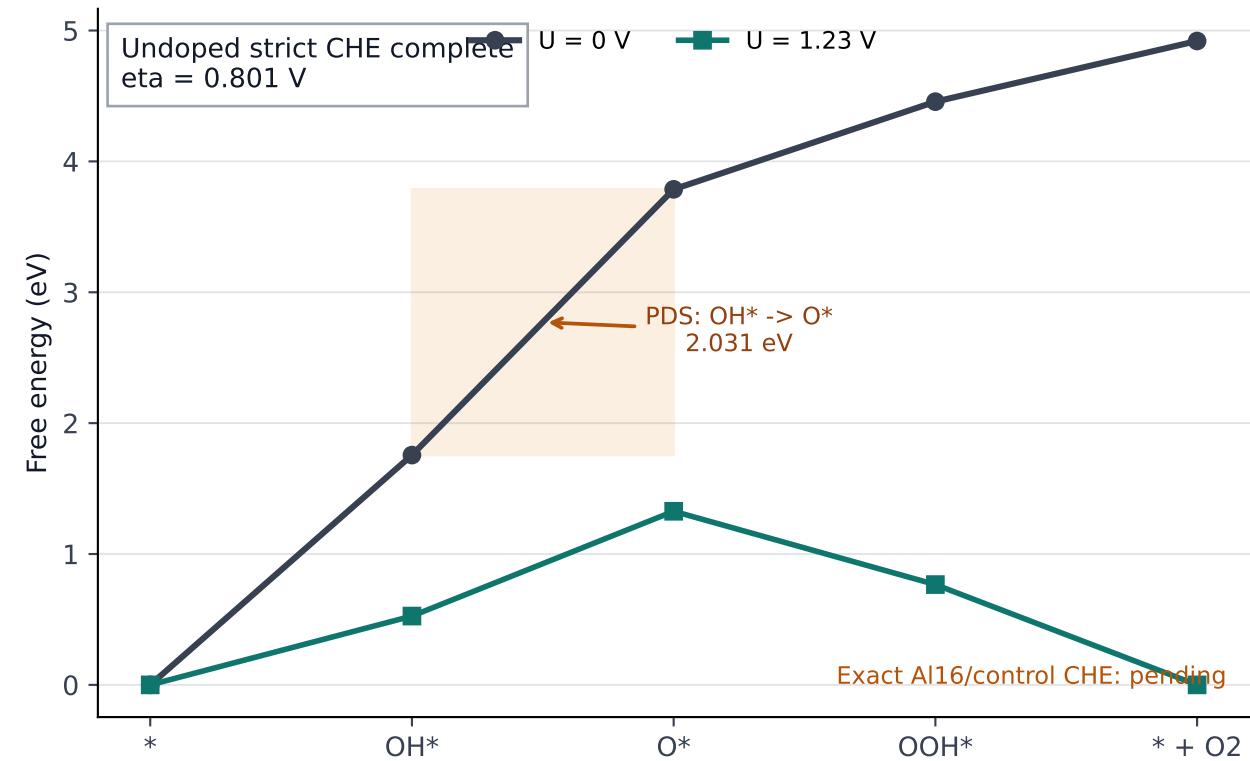


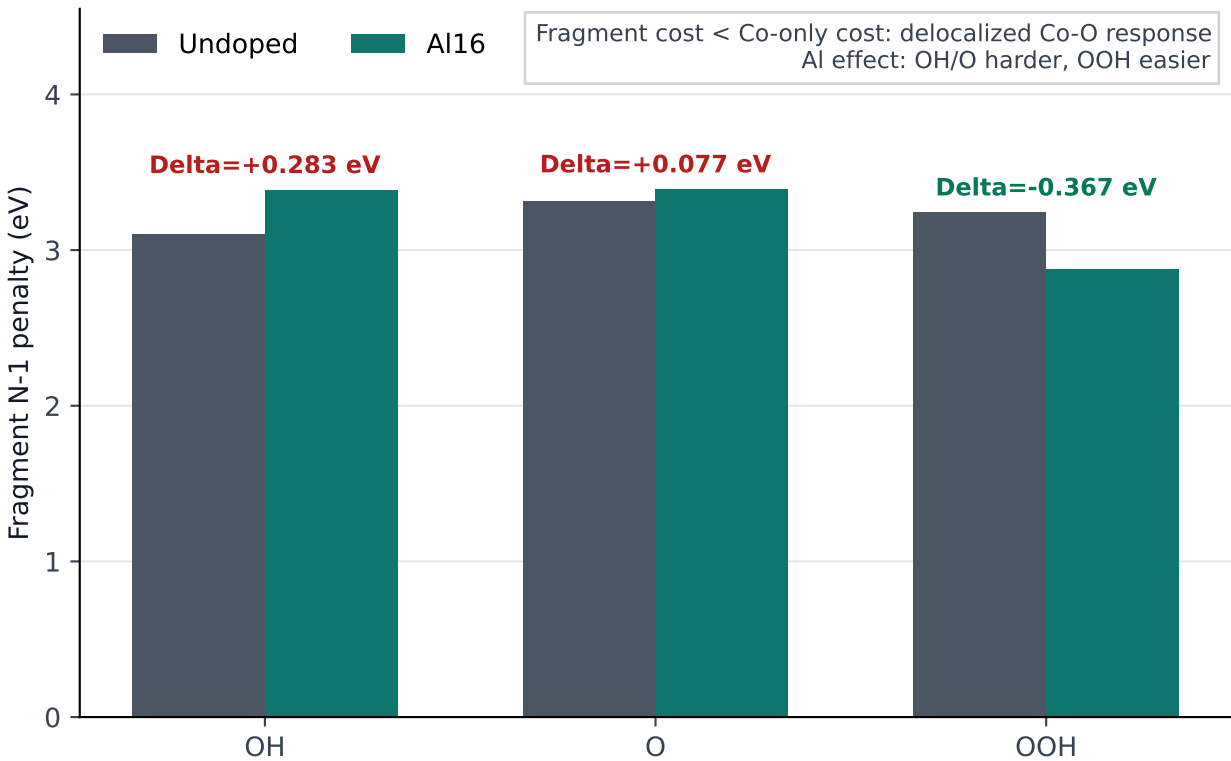
CoOH OER with Al16: interim mechanistic summary

Validated diagnostics through 2026-08-19 | Final matched Al16/control CHE remains pending

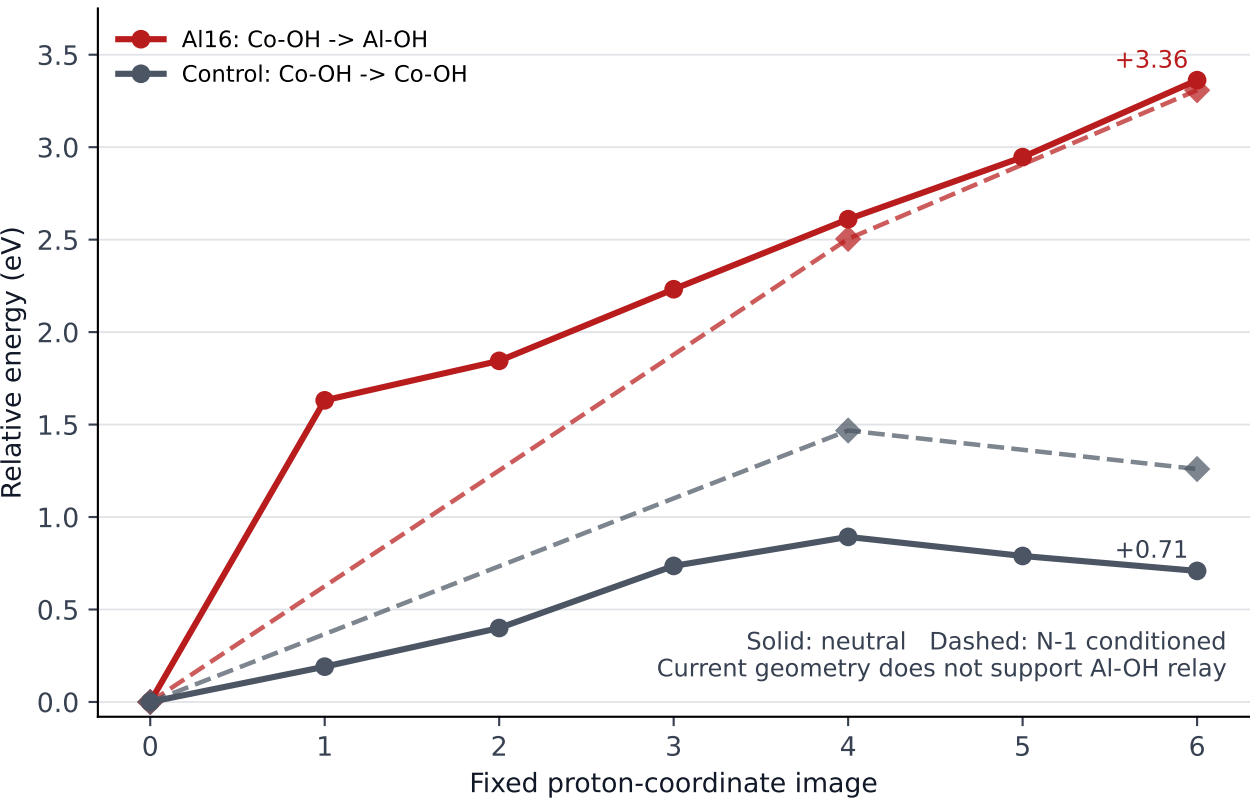
A Conventional four-step AEM/CHE



B cDFT oxidation localization



C Adjacent-OH proton relay screen



D Electronic diagnostics and workflow

Active Co net charge	Active Co moment	Occupied Co-d center
-0.027 e	+0.146 μ_B	+0.082 eV
slightly less positive	higher local spin	upshift vs own EF
cDFT matched scans		
14/14 valid		
Proton-relay screen		
14 neutral + 6 N-1		
Fixed-geometry VASP		
3/8 complete		
Bader analyses		
3/8 valid		
Matched electronic pairs		
1/4 pairs		
Strict Al16/control relax		
8 running; 0 valid final		
Exact matched CHE		
pending relax/statics/vib		

Bare result: weak electronic tuning, not strong Co electron withdrawal.
COHP pending licensed LOBSTER; structural effects await final relaxed statics.

N-1 means one electron removed from the constrained fragment. Bader charges are partition-dependent diagnostics, not formal oxidation states.